CS3003: IT Project Management

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Syllabus Outline

1. **Project Management (PM) Fundamentals**
2. **Business Case and Investment Appraisal, Common Cost elements in IT projects; Models of S/W acquisition/licensing. Basics of Procurement and Contracts Management.**
3. **Project Organization** (e.g. Project Governance Structure; Stakeholders Analysis; Role of Project Manager)
4. **Basic Processes of a Project. References to relevant industry standards, such as PMI, ISO 21500 and Prince2.**
5. **Project Planning**
6. **Project Monitoring, Control and Implementation Processes**
7. **Fundamentals of Team Management**
8. **Project Risk Management:** fundamental concepts and terminology; the risk management process.
9. **Quality Aspects**
10. **The IT Project Manager Skills, Profession and Code of Ethics**: introduction and outline of relevant topics.

Learning Outcomes

- LO1 - **Understand the fundamental concepts, terminology and cost elements of IT Projects.**
- LO2 - **Apply basic Project Planning techniques.**
- LO3 - **Explain the procedures needed to monitor, control and report upon an IT Project.**
- LO4 - **Demonstrate an understanding of the steps needed to build/maintain effective development teams.**
- LO5 - **Discuss and where appropriate apply the principles of Project Risk Management.**
- LO6 - **Explain the ways in which appropriate quality attributes of the products of an IT Project can be assessed and assured.**

Skills

- S1 - **Problem Solving and Critical Thinking.**
- S2 - **Written Communication/Argumentation.**
- S3 - **Presentations skills.**

Reading Resources

1. Schwalbe, Kathy. *Information Technology Project Management*. Course Technology, (8th Ed), 2015, ISBN: 1285452348.
2. Field, Mike and Keller, Laurie. *Project Management*. Open University, International Thomson Press, 1998, ISBN: 1861522746.
3. Hughes, Bob and Cotterell, Mike. *Software Project Management*. McGraw-Hill (5th Ed), 2009, ISBN: 0077122798.
4. Brooks, Frederick. *The Mythical Man-month: Essays on Software Engineering*. Addison Wesley (2nd Ed), 1995, ISBN: 0201835959.
5. DeMarco, Tom and Lister, Tim. *Peopleware: Productive Projects and Teams*. Addison Wesley (3rd Ed), 2013, ISBN: 0321934113.

Additional readings and handouts will be provided on specific topics, e.g. relevant material from the Prince2 Manual and the PMBOK Guide, as well as papers and web resources.

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Part 1. Project Management Fundamentals

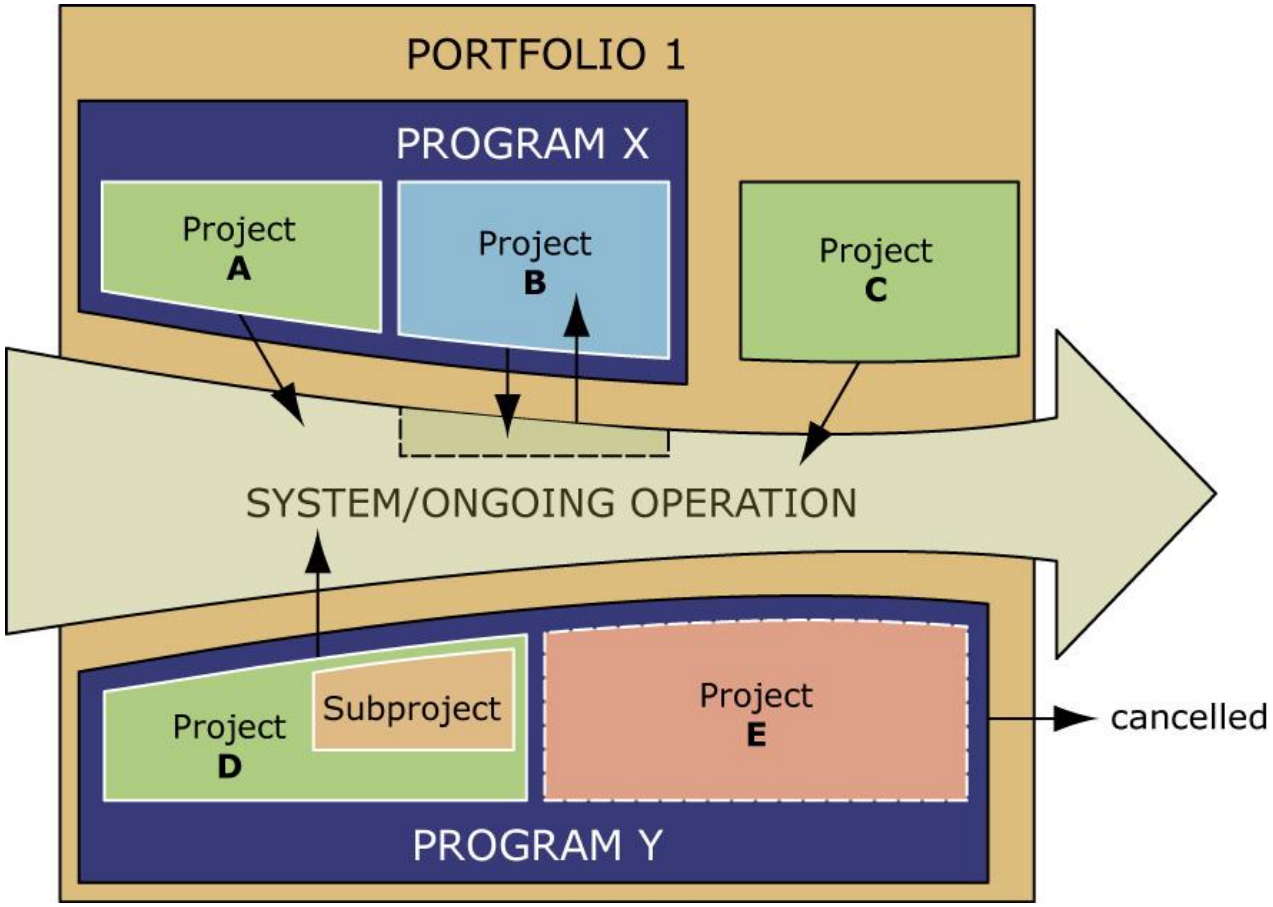
What is a “Project”?

Entity/”Framework”	Definition
PMI, PMBOK Guide	*“A project is a temporary endeavour undertaken to create a unique product, service, or result.”*
APM	*“A project is a unique, transient endeavour, undertaken to achieve planned objectives, which could be defined in terms of outputs, outcomes or benefits.”*
ISO 21500:2012	*“A project consists of a unique set of processes consisting of coordinated and controlled activities with start and end dates, performed to achieve project objectives.”*
Prince2	“*A temporary organisation that is needed to produce a unique and predefined outcome or result at a prespecified time using predetermined resources”* or “*A management environment that is created for the purpose of delivering one or more business products according to a specified Business Case”*
Agile	?

Some Key Project Characteristics

- Driven by business (organization) needs and have concrete objectives/results.
- Temporary, have a definite Beginning and End.
- Consume “Resources”.
- Involve Risk.
- Have defined Roles and Responsibilities.
- Have Visibility.

Projects, Programmes and Portfolios



- **Programme**: a group of related projects managed in a coordinated way. Examples?
- **Portfolio Management**: selection, prioritisation and control of an organisation's projects/programmes according to its strategic objectives and ability to deliver.

Project Management & Project Lifecycle

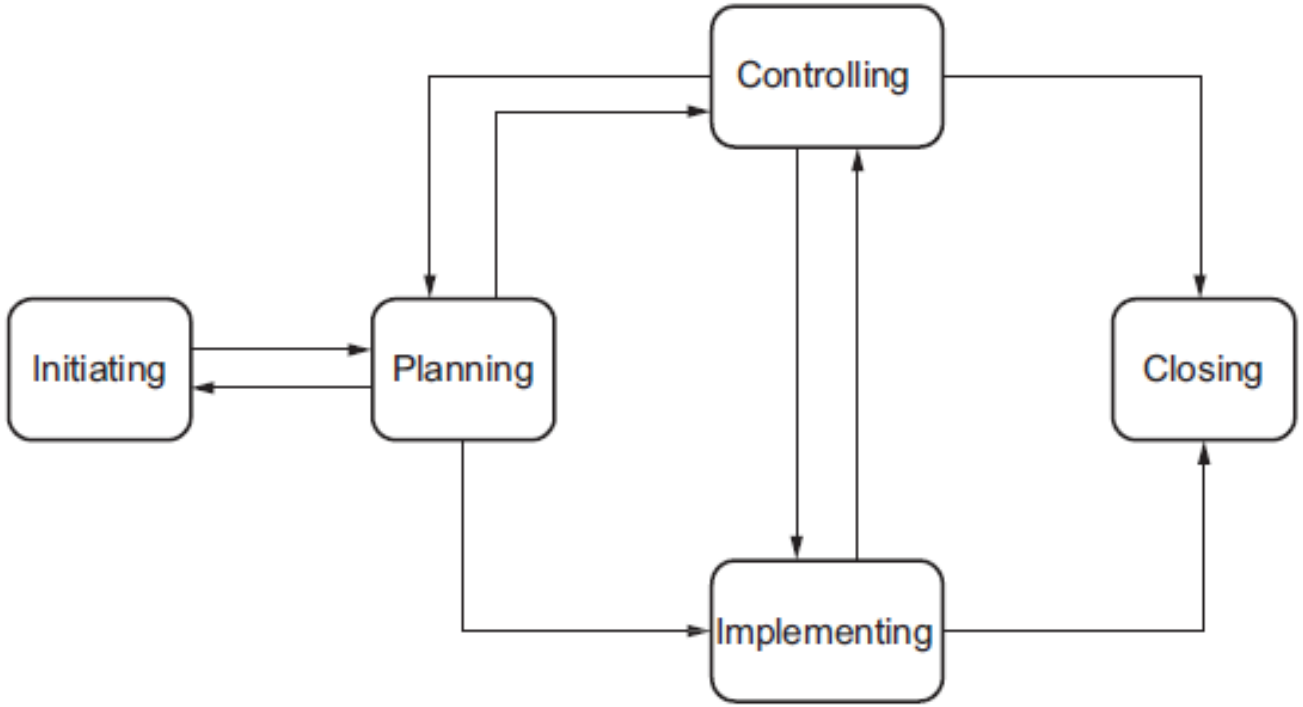
- **Project Management**: Various definitions, such as "*the application of processes, methods, knowledge, skills and experience to achieve the project objectives*".

- **Project Processes**: "*activities to be carried out during the project*" or "*sequences of stages that each project goes through*". The processes should be managed through Project Management.

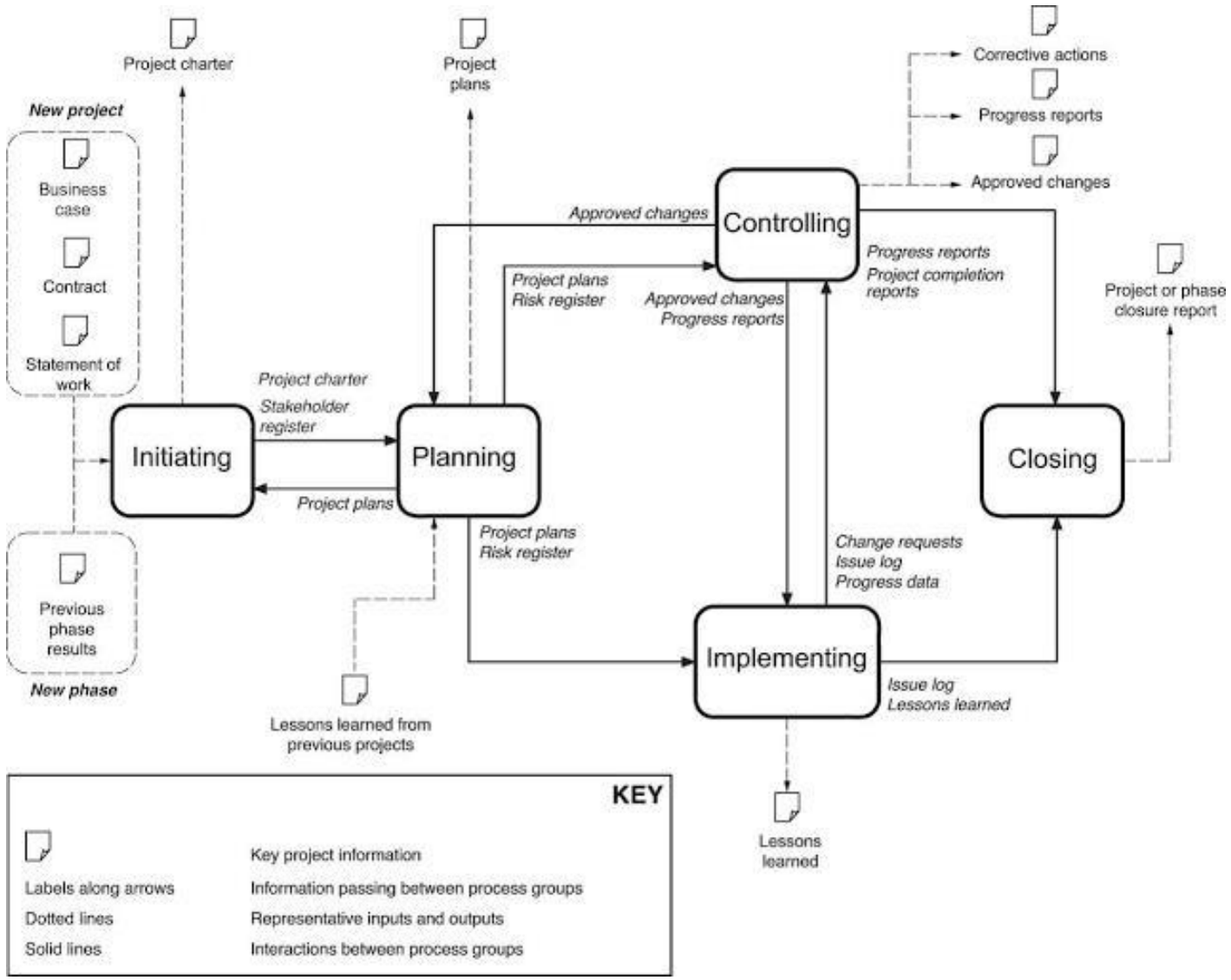
Project Management & Project Lifecycle

Main Processes/Process Groups of a Project (from ISO 21500 & PMI)

- Initiating
- Planning
- Controlling (Monitoring)
- Implementing (Executing)
- Closing



Key Project Process Groups (ISO 21500)



Key Elements of a Project

Business Case

Scope

Budget/Costs

Plan

Governance Structure

Project Methodology

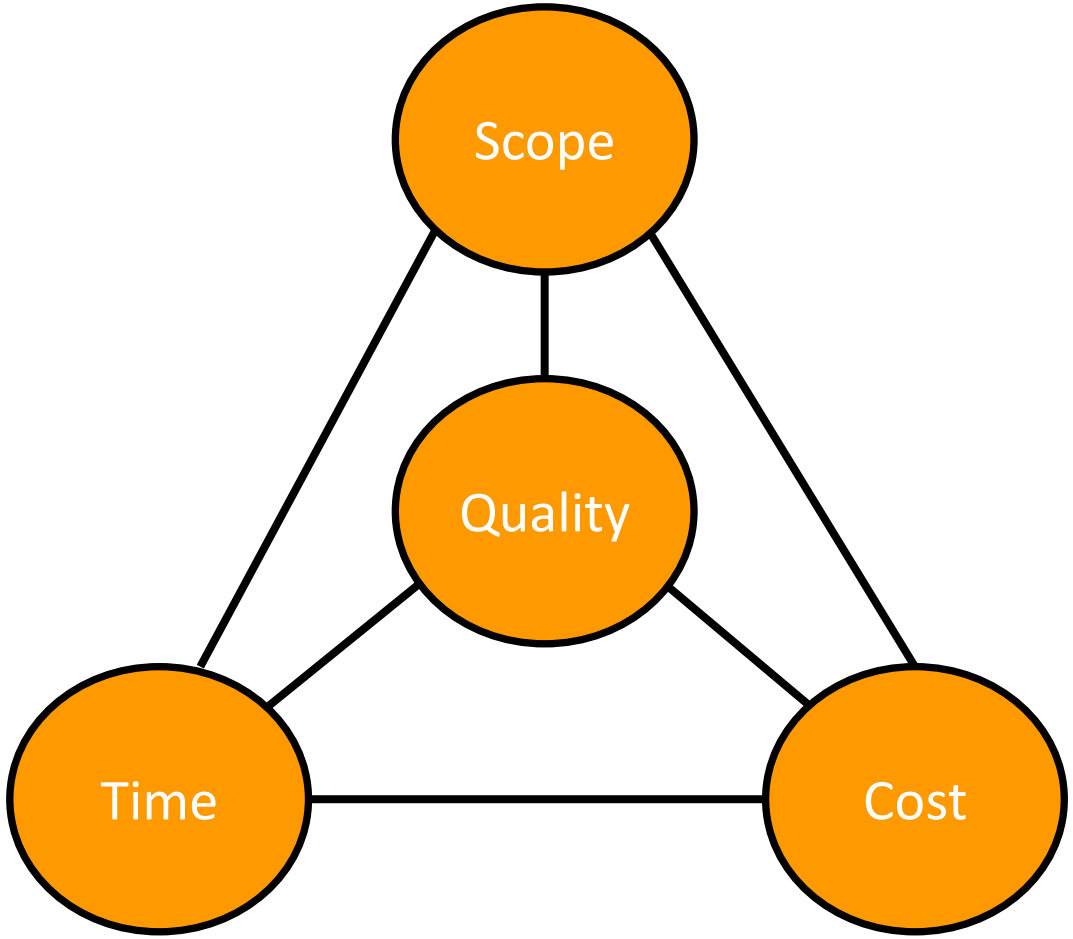
Stakeholders

Resources

Project Team

Project Manager

Project Variables aka “Project Triangle” or “Triple” or “Quadruple” Constraint



Main distinguishing characteristics of IT Projects

1. The end-product is intangible

The end-product (e.g. software) is harder to be seen ("touched"). IT Project Managers and other stakeholders cannot see progress by simply looking at the artefact that is being constructed.

2. Degree of Uniqueness

Most software projects are usually different in a number of ways from previous projects. Even managers who have significant experience may find it difficult to anticipate issues. IT Systems are extremely human-intensive and humans are non-linear systems!

3. Degree of Uncertainty/Change

Requirements of most IT projects change during the project due to a number of reasons (e.g. business needs, market changes)

4. Degree of inter-dependencies

IT Systems rarely live in isolation and there exists a significant degree of interdependencies to other systems/overall IT environment. A change or defect may trigger a cascade of effects.

5. Communication/ Common Terminology

There is often a lack of a common vocabulary between IT and Business people, and even between IT people, mainly due to the proliferation of terms, lack of standardization, the pace of technology evolution, inter-alia.

Some Common “Misconceptions”

- Project Management BoK Vs “Methodologies”/“Frameworks”
- Descriptive Vs Prescriptive
- Defined Vs Empirical/Incremental Approaches
- IT Project Management Vs Software Development & Process Models

➢ **PMI** Vs **APM** Vs **Prince2** Vs **Agile** (e.g. Scrum, Kanban, XP, AgilePM, LESS, SAFe)

We will revisit these at a later stage!